

MIDTJYLLANDS AP, Denmark

WMO: 060600

Lat: 56.293N

Lon: 9.114E

Elev: 52

StdP: 100.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -7.3	(d) -11.7	(e) 1.4	(f) -8.4	(g) -9.4	(h) 1.7	(i) -5.7	(j) 14.7	(k) 6.1	(l) 12.9	(m) 6.3	(n) 1.9	(o) 20	(p) 0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.2	(c) 27.5	(d) 18.8	(e) 25.5	(f) 17.9	(g) 23.5	(h) 17.1	(i) 20.0	(j) 24.7	(k) 19.0	(l) 23.6	(m) 18.0	(n) 21.8	(o) 4.3	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.3	21.4	17.3	12.4	20.5	16.4	11.8	19.6	57.3	24.8	54.1	23.8	51.0	21.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.9	8.8	-13.5	30.3	2.8	1.6	-15.5	31.5	-17.1	32.4	-18.7	33.3	-20.7	34.5
			-13.8	21.9	2.7	1.2	-15.7	22.7	-17.3	23.4	-18.9	24.1	-20.9	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	1.1	1.3	3.3	7.4	11.3	14.2	16.8	16.8	13.4	9.3	5.2	2.3
	DBStd	6.65	4.17	3.93	3.54	3.46	3.34	2.86	2.92	2.77	2.65	3.24	3.48	4.44
	HDD10.0	1285	277	242	209	93	24	1	0	0	3	50	147	239
	HDD18.3	3606	535	476	467	328	221	128	67	66	148	279	394	498
	CDD10.0	766	0	0	1	14	63	128	211	211	106	29	2	0
	CDD18.3	46	0	0	0	0	1	5	21	17	2	0	0	0
	CDH23.3	446	0	0	0	2	21	67	204	144	9	0	0	0
	CDH26.7	84	0	0	0	0	1	10	45	28	1	0	0	0
(14) Wind	WSAvg	4.2	4.7	4.7	4.7	4.3	4.2	4.3	3.8	3.7	3.9	4.1	4.2	4.3
(15) Precipitation	PrecAvg	828	76	58	52	45	49	64	68	83	83	94	76	80
	PrecMax	1014	124	129	102	86	107	124	126	141	169	205	166	169
	PrecMin	518	6	17	6	5	3	4	17	20	31	26	26	20
	PrecStd	116	33	30	25	24	24	30	29	31	38	40	33	40
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	10.2	14.3	21.8	25.8	28.1	30.3	29.2	24.6	18.3	14.0	11.2
		MCWB	8.7	8.8	9.1	14.0	17.1	18.1	19.8	19.4	17.6	15.1	13.0	10.4
	2%	DB	8.3	8.7	11.4	18.7	23.0	25.3	27.8	27.0	21.9	16.2	12.2	10.2
		MCWB	7.6	7.8	7.8	11.8	15.5	17.6	19.0	18.9	16.6	14.1	11.2	9.6
	5%	DB	7.4	7.6	9.5	16.2	20.5	22.7	25.8	24.7	19.7	15.0	11.1	9.0
		MCWB	6.6	6.9	7.2	10.9	14.0	16.2	18.3	18.0	15.7	13.5	10.1	8.3
	10%	DB	6.6	6.7	8.1	13.7	18.1	20.2	23.4	22.3	18.0	13.9	10.0	8.1
		MCWB	5.8	5.9	6.5	9.6	12.9	15.2	17.2	17.3	14.9	12.6	9.2	7.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	9.1	10.3	15.1	18.8	20.3	21.2	21.5	19.2	16.0	13.1	10.8
		MCDB	9.3	9.5	12.7	20.0	24.4	25.3	27.3	25.8	22.6	17.3	13.7	11.3
	2%	WB	7.8	7.9	8.6	12.6	16.5	18.6	20.1	19.8	17.3	14.8	11.5	9.6
		MCDB	8.2	8.4	10.3	17.0	21.0	23.1	25.5	24.0	20.2	15.8	12.2	10.0
	5%	WB	6.8	6.9	7.6	11.3	15.0	17.0	19.1	18.9	16.4	13.7	10.3	8.5
		MCDB	7.2	7.4	9.0	15.1	19.3	21.4	24.1	23.0	18.8	14.6	10.9	8.9
	10%	WB	5.9	5.9	6.7	10.1	13.5	15.7	17.9	17.9	15.5	12.8	9.2	7.5
		MCDB	6.4	6.4	7.8	12.9	17.2	19.5	21.8	21.4	17.6	13.8	9.9	8.0
(35) Mean Daily Temperature Range	MDBR		4.7	4.9	6.7	8.9	9.4	8.8	9.2	8.9	7.7	6.3	4.9	4.6
	5% DB	MCDBR	4.4	4.7	8.7	13.2	13.7	13.4	13.6	12.5	10.1	6.7	4.7	4.2
		MCWBR	4.4	4.1	5.8	7.5	7.5	7.0	6.3	5.6	5.8	5.0	4.4	4.3
	5% WB	MCDBR	4.4	4.5	7.3	11.6	11.8	11.5	12.0	10.6	8.6	5.9	4.7	4.3
		MCWBR	4.5	4.1	5.4	7.0	7.2	6.8	6.2	5.5	5.5	4.8	4.5	4.3
(40) Clear-Sky Solar Irradiance	taub		0.314	0.334	0.355	0.390	0.382	0.376	0.392	0.393	0.370	0.349	0.326	0.300
	taud		2.221	2.278	2.300	2.257	2.326	2.364	2.337	2.345	2.395	2.384	2.291	2.215
	Ebn at Noon		565	717	801	825	857	866	842	814	780	695	550	482
	Edh at Noon		56	78	97	117	116	113	114	106	88	70	52	44
(44) All-Sky Solar Radiation	RadAvg		0.46	1.05	2.40	3.99	5.23	5.63	5.16	4.09	2.81	1.44	0.57	0.33
	RadStd		0.08	0.15	0.22	0.48	0.50	0.46	0.58	0.33	0.30	0.17	0.09	0.04

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d) +0.38	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) -150	(l) N/A	(m) N/A
(47) Regional (34 neighbors)		+0.36	N/A	N/A	N/A	N/A	N/A	-123	-167	N/A	N/A

Nomenclature: See separate page